

What's the lift: Predicting Promotion Impact for Smarter Replenishment

Company Sponsor: Estefania Elizondo, Ximena Zuniga, Alexandra Lillback
Faculty Advisor: Stephen C Graves



Brian Lu



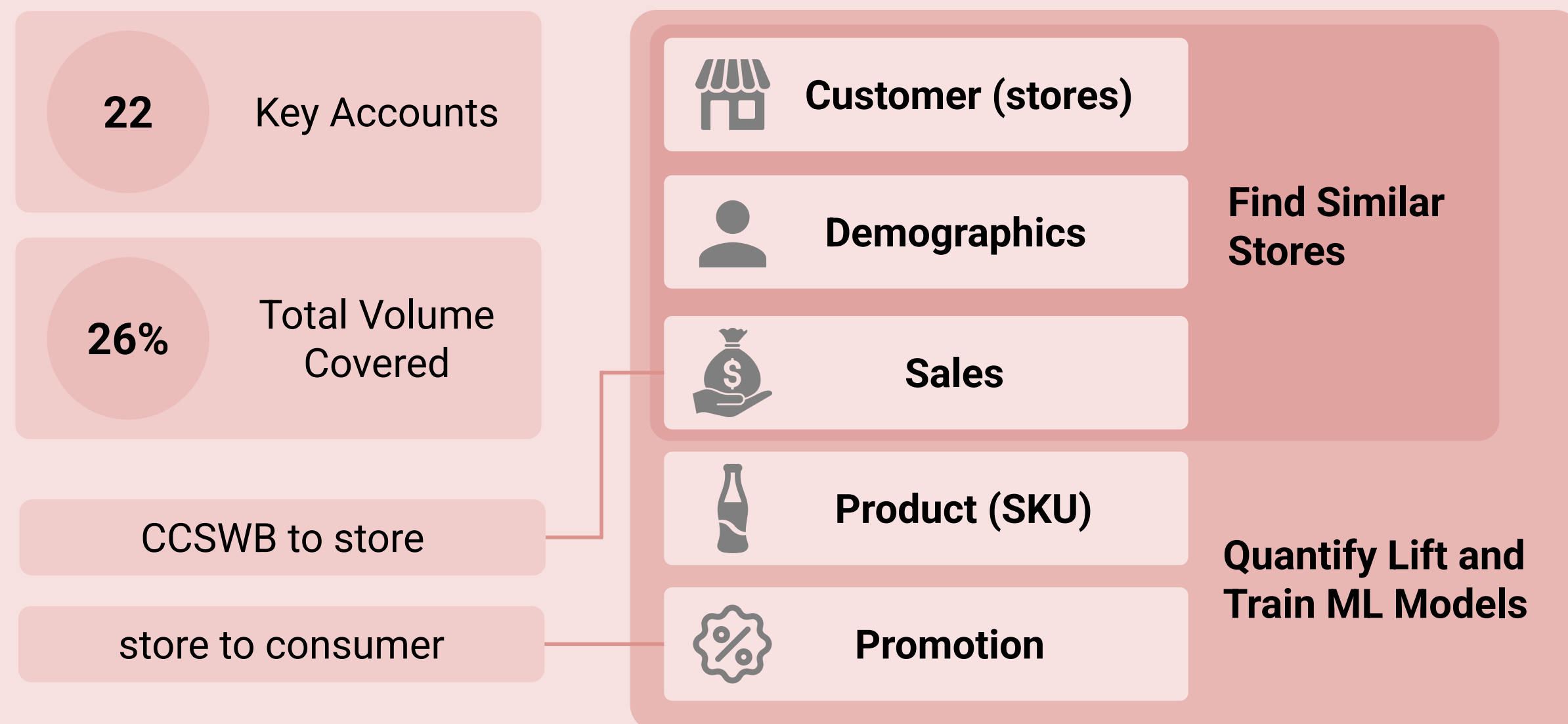
Annie Liu

Overview

Problem Statement

- CCSWB is the 3rd largest Coca-Cola bottler in the U.S., supplying over 1,500 beverage products to more than 88,000 retail locations. Weekly replenishment decisions are supported by a machine learning-based Suggested Order (SO) system
- Replenishers are frontline employees responsible for final replenishment decisions. During each store visit, they restock shelves, assess inventory and shelf conditions, and determine how much of each product should be ordered
- Today, the SO system has no awareness of when a promotion is running. Replenishers rely on separate promotion planning sheets and manually estimate additional order quantities

Data



How can we help replenishers make better ordering decisions during promotions?

Methodology

There was no existing module to quantify the sales effect (i.e., lift) of historical promotions. The lift is confounded by seasonality, baseline demand, and store-level variation, making it hard to isolate the true impact of a promotion from all other sales-driving factors.

Therefore, our solution is a 2-step process:

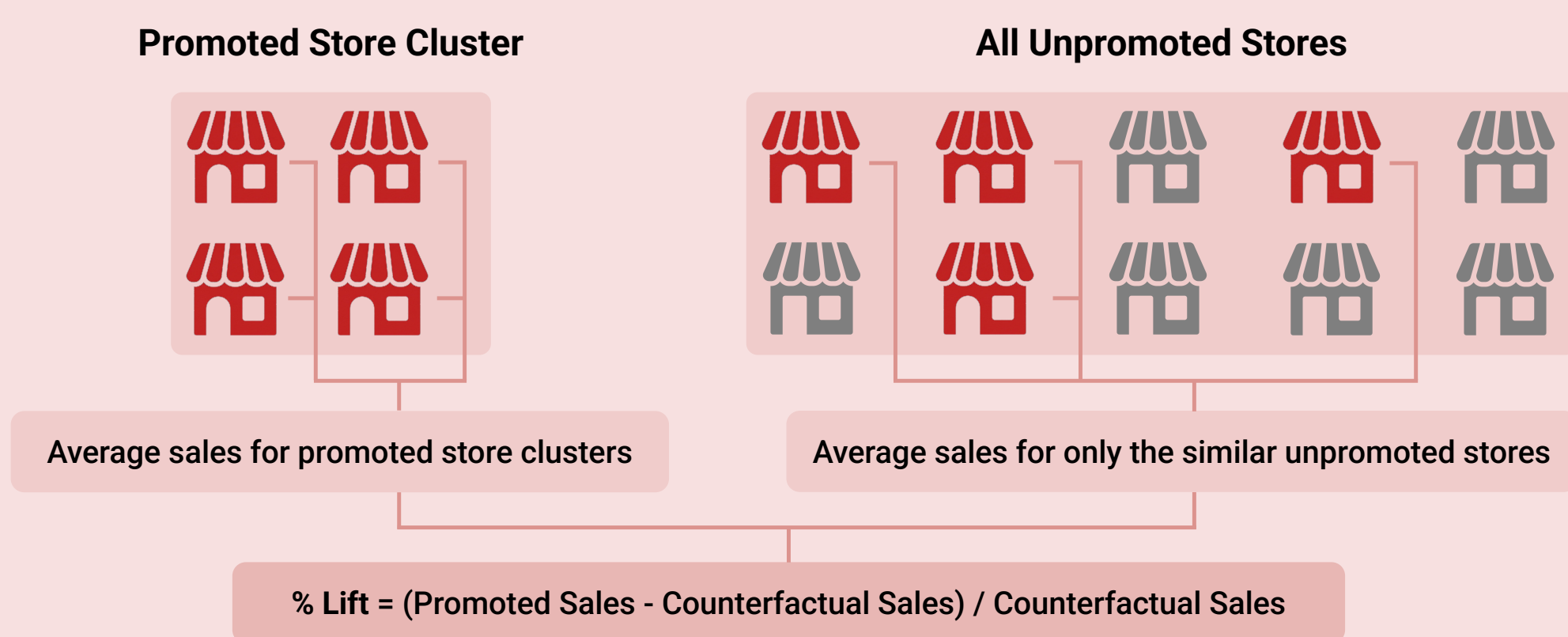
Step 1: Build a lift quantification module using historical promotions

Step 2: Train a lift prediction model for future promotions

Step 1: Quantify Lift

Lift Quantification Module

The no-promotion outcome is never observed. It is estimated from similar unpromoted stores, and promotion lift is calculated against this baseline.



Sample Lift (2026-05-27 → 2026-06-02)



Step 2: Predict Lift

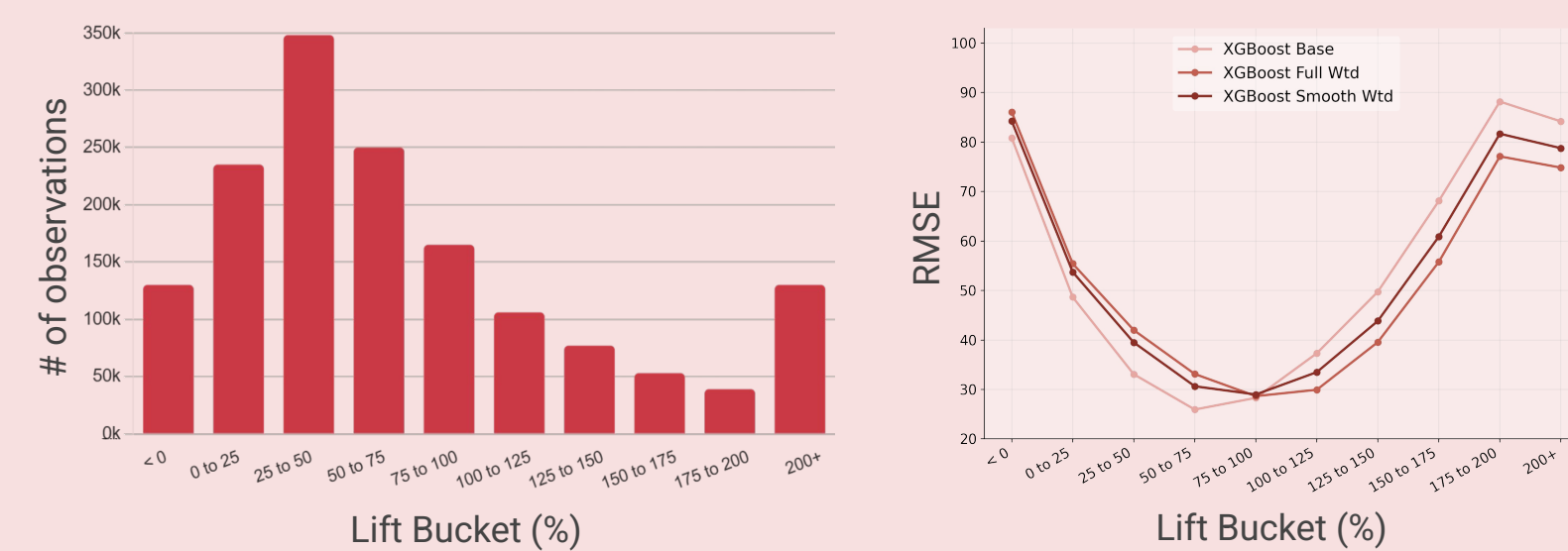
Feature Engineering

To capture the factors that influence promotional lift, additional features were created for each store-product pair. Key features include:

- Average historical promotion lifts
- Historical promotion frequency
- Seasonal timing of promotions

Smooth Weighted XGBoost

Lift buckets with fewer observations are harder to predict, so a **smooth inverse-frequency weighting** scheme was applied during model training.



Result

The final prediction model achieved a **test MAE of 43.7%**. The predicted lift closely matched the overall trend of the actual lift across products and customers.

Due to regularization, predictions **were smoother over time** while still capturing the overall trend.



Impact

Coverage

25%
Total Volume

Promotions drive 25% of order volume, and this model targets exactly where the opportunity is

Revenue Captured

+400k
USD per week

Compared with current ordering behavior

+677k
USD per week

Compared with output of Suggested Order



CCSWB Suggested Order: Promotion Lift Forecasting Model

Final Project Presentation

Brian Lu, Annie Liu

CCSWB Sponsors: Estefania Elizondo, Ximena Zuniga, Alexandra Lillback

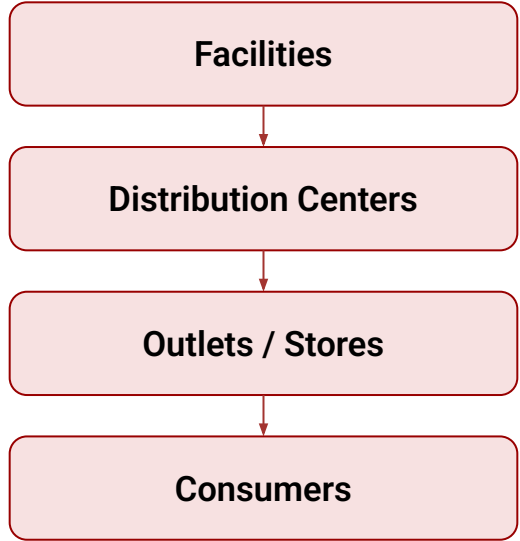
Faculty Advisor: Stephen C Graves

► Who is Coca-Cola Southwest Beverages (CCSWB)

Coca-Cola v.s. CCSWB



Coca-Cola Lifecycle



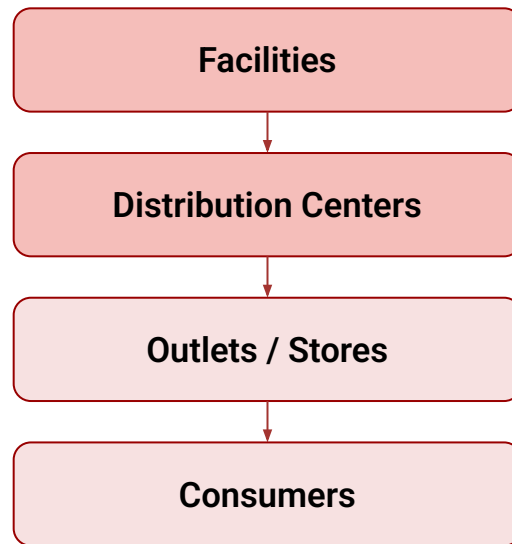
► Who is Coca-Cola Southwest Beverages (CCSWB)

Coca-Cola v.s. CCSWB



- 3rd largest Coca-Cola bottler in the U.S.
- Serves TX, NM, OK, and AR
- Over 1,500 products across 88,000 retail locations

Coca-Cola Supply Chain



► Meet Our Frontline Heroes: Replenishers

What Replenishers Do

- Visit multiple stores and restock products
- Assess inventory and demand
- Place orders for each store

Challenges They Face

- Hundreds of ordering decisions every day
- Promotions related decisions are time consuming
- Current Suggested Order system underperforms



► Problem Statement

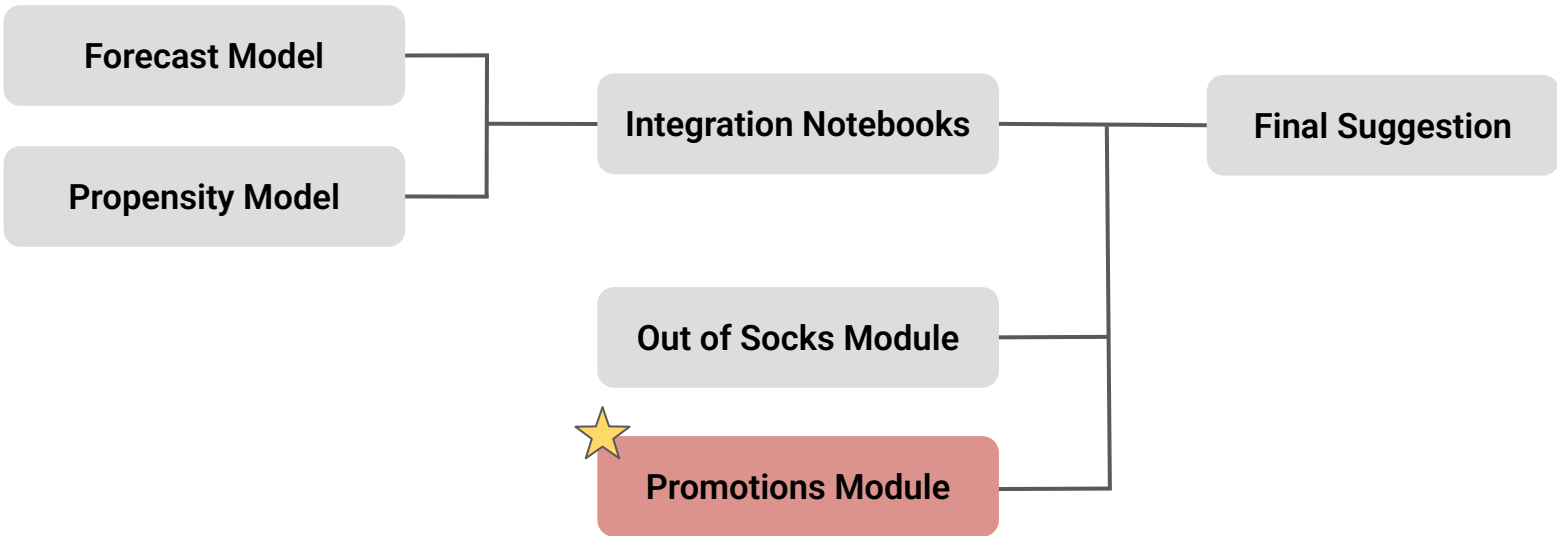
Make replenishers' work easier

How can we help replenishers make better ordering decisions during promotions?



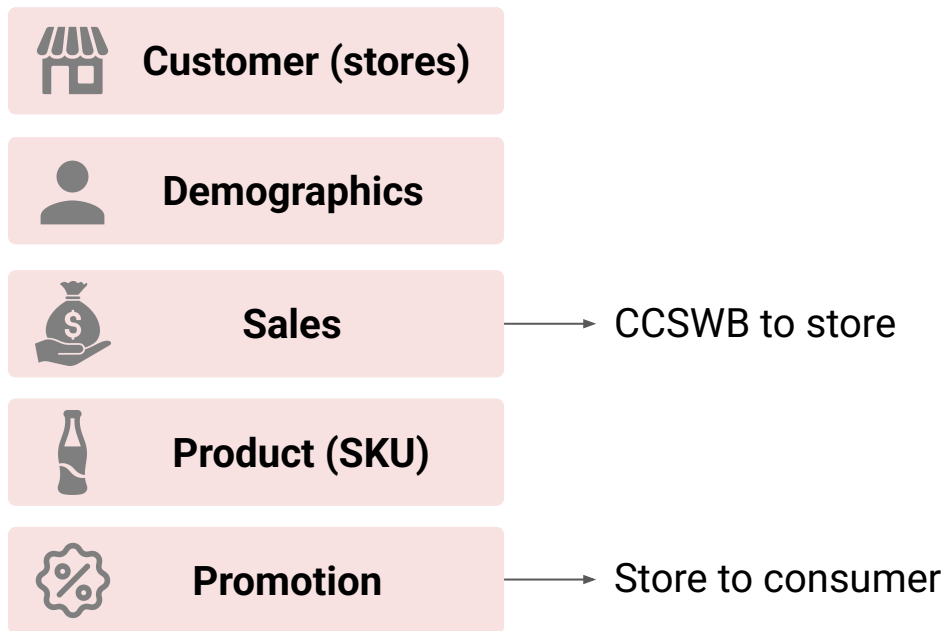
► Project Overview

Promotion Lift Model and the full Suggested Order pipeline:



► The Data

Five rich datasets are used to identify similar stores and predict promotion lift



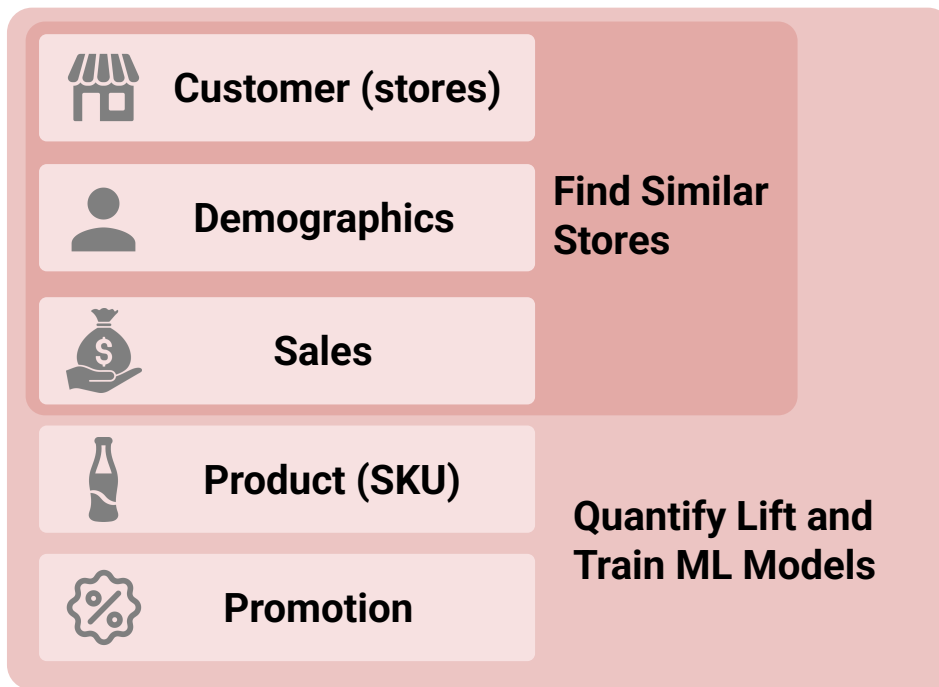
► The Data

Five rich datasets are used to identify similar stores and predict promotion lift



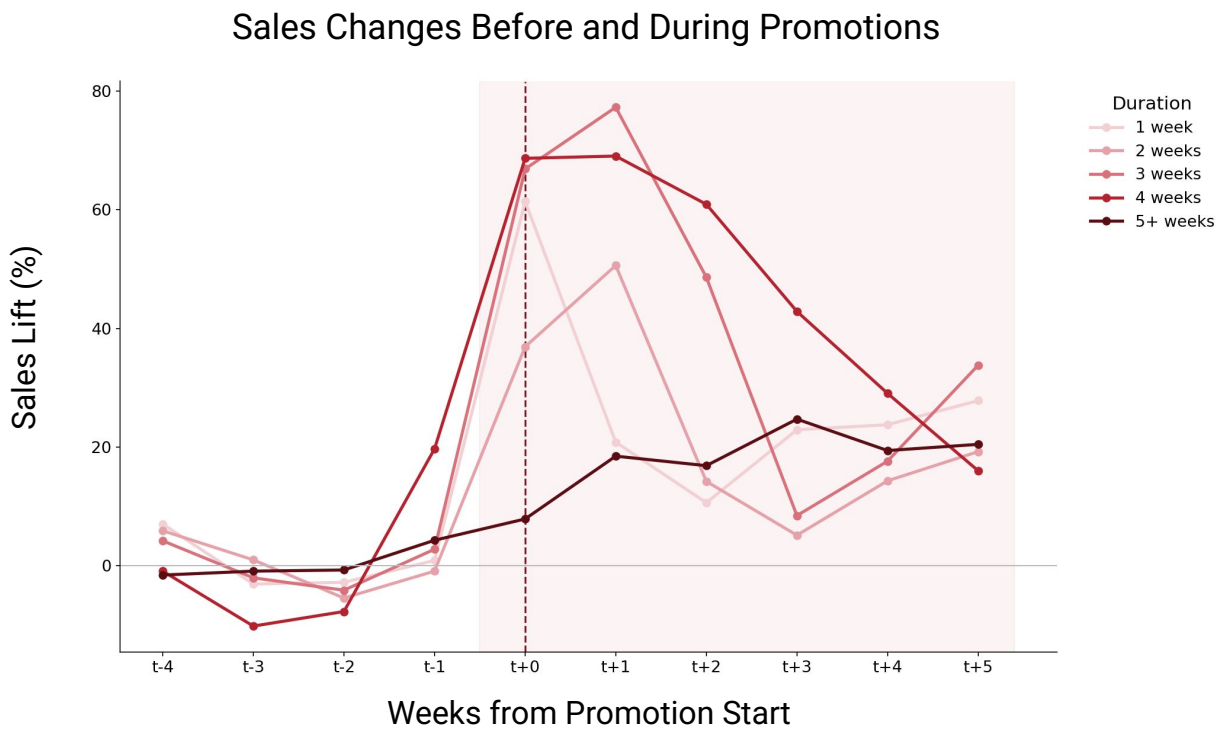
► The Data

Five rich datasets are used to identify similar stores and predict promotion lift



► Promotions Drive Clear Sales Lift

Promotions increase sales immediately, and the effect lasts throughout the promotion



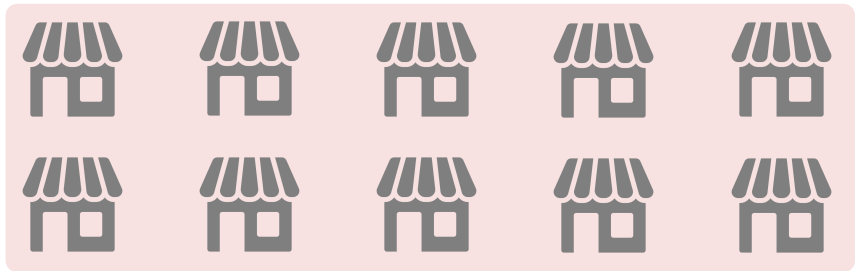
► Lift Quantification

Compare promoted stores with similar unpromoted stores to estimate lift

Promoted Store Cluster



All Unpromoted Stores



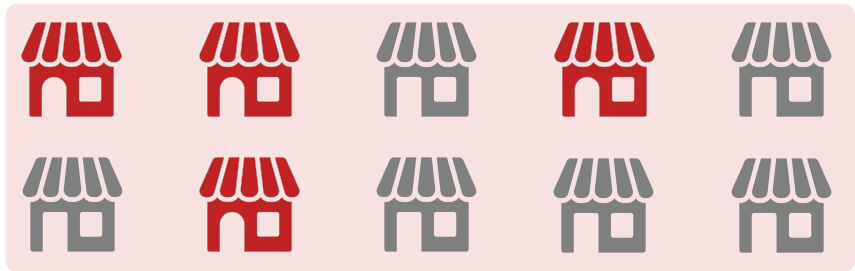
► Lift Quantification

Compare promoted stores with similar unpromoted stores to estimate lift

Promoted Store Cluster



All Unpromoted Stores

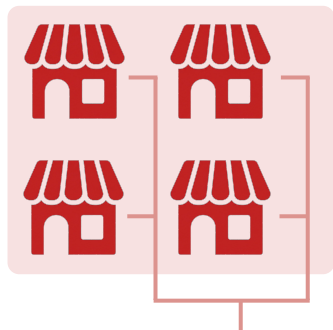


Identify similar stores

► Lift Quantification

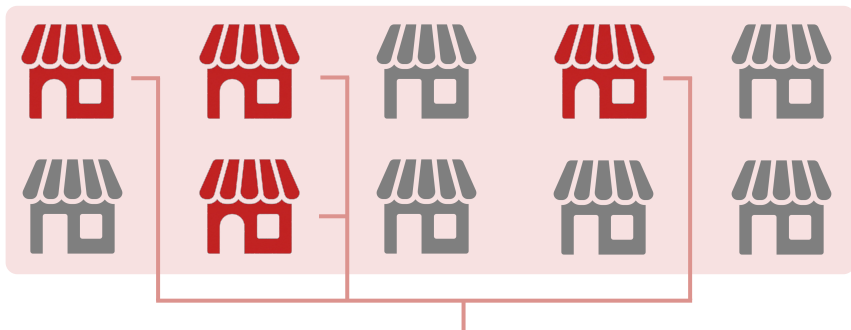
Compare promoted stores with similar unpromoted stores to estimate lift

Promoted Store Cluster



Average sales for promoted store clusters

All Unpromoted Stores



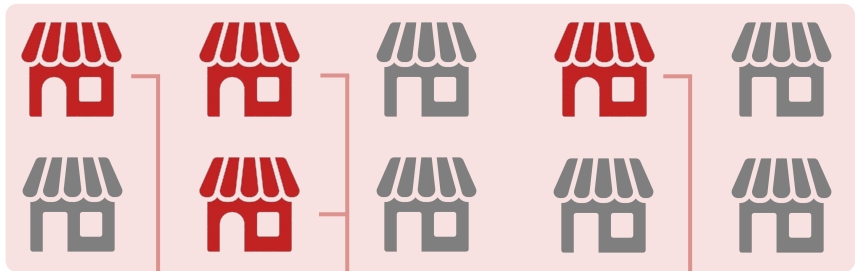
Average sales for only the similar unpromoted stores

► Lift Quantification

Compare promoted stores with similar unpromoted stores to estimate lift

Promoted Store Cluster

All Unpromoted Stores



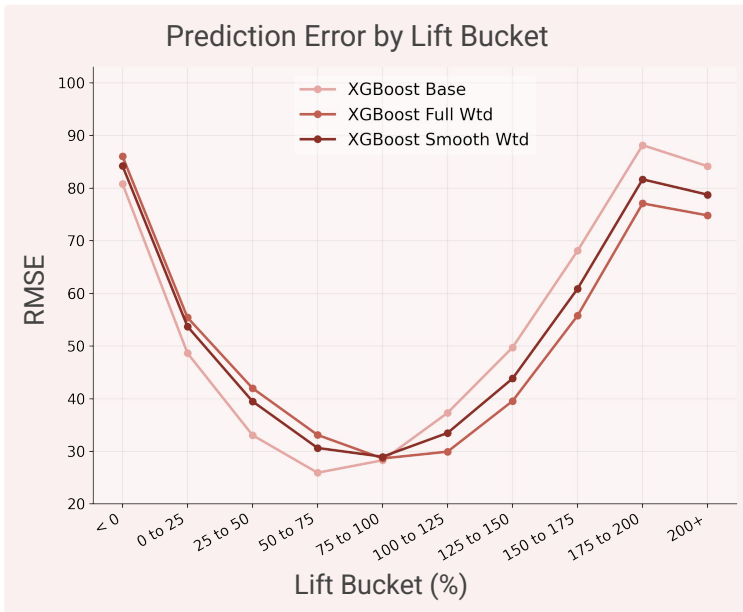
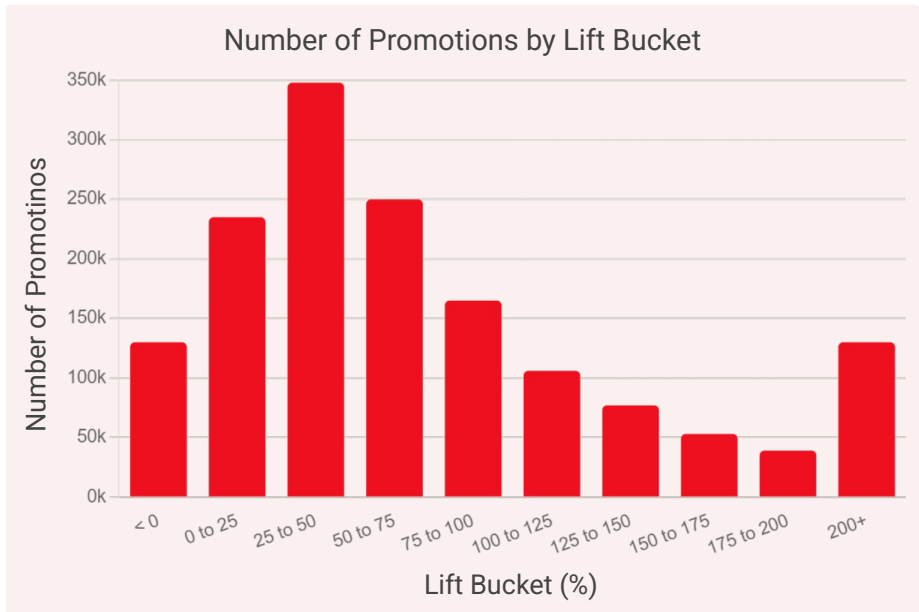
Average sales for promoted store clusters

Average sales for only the similar unpromoted stores

$$\% \text{ Lift} = (\text{Promoted Sales} - \text{Counterfactual Sales}) / \text{Counterfactual Sales}$$

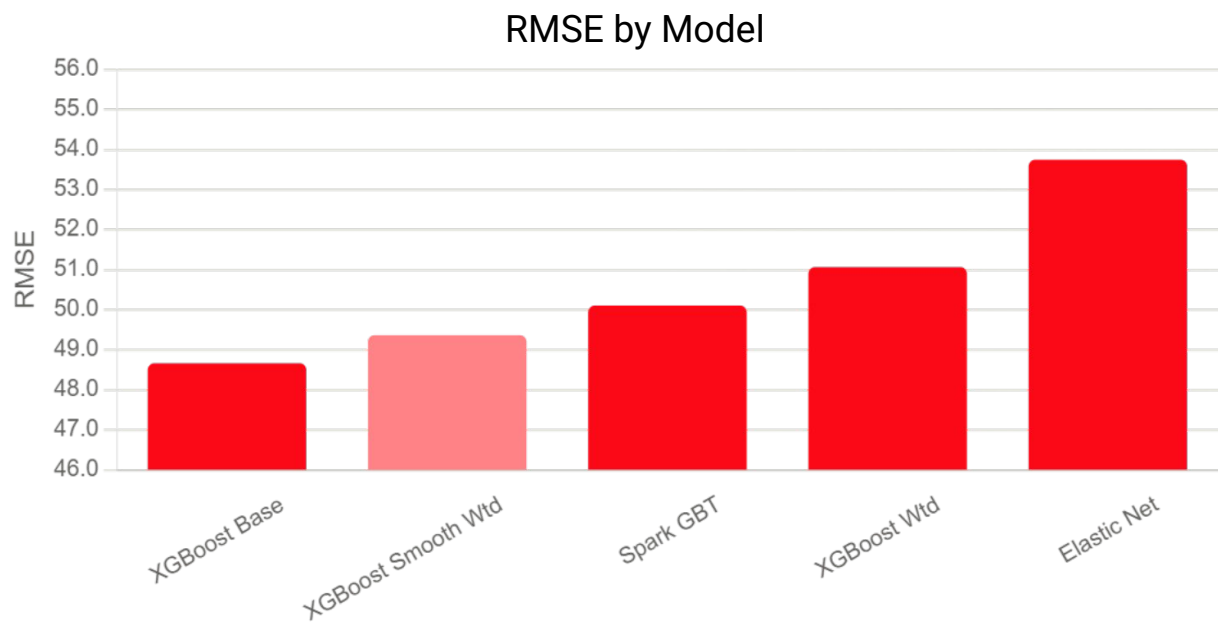
► Weighted Training for Rare Promotions

Less training data leads to higher prediction error, motivating the use of weighted training



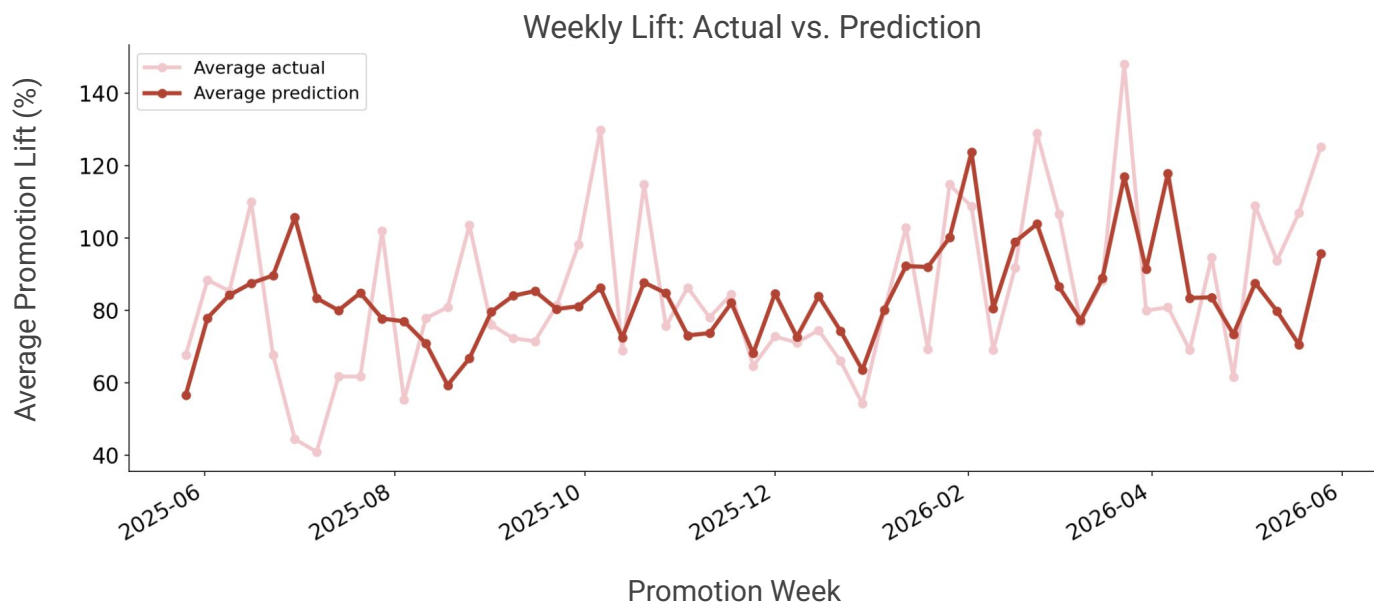
► Training Result and Model Selection

Although Base XGBoost achieved the lowest overall RMSE, **Smooth Weighted XGBoost** was selected for its better performance on high-lift promotions



► Predictions Capture Overall Lift Patterns

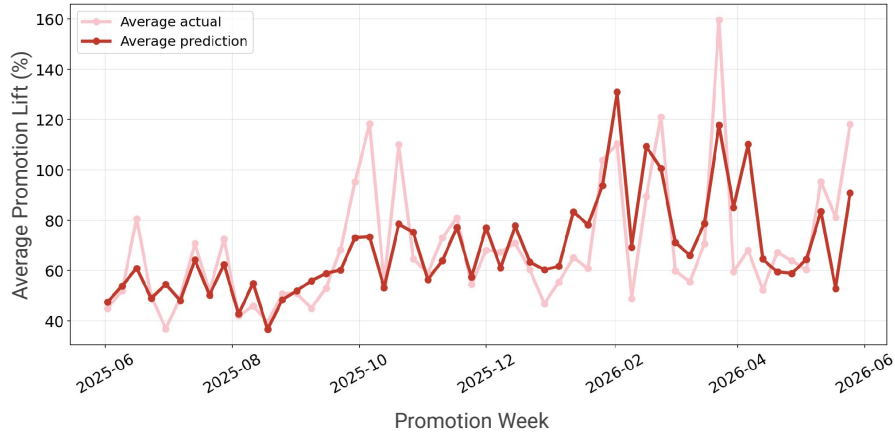
Time-series comparison showed that the model **captured actual lift patterns**, while regularization prevented extreme predictions.



► Predictions Capture Overall Lift Patterns

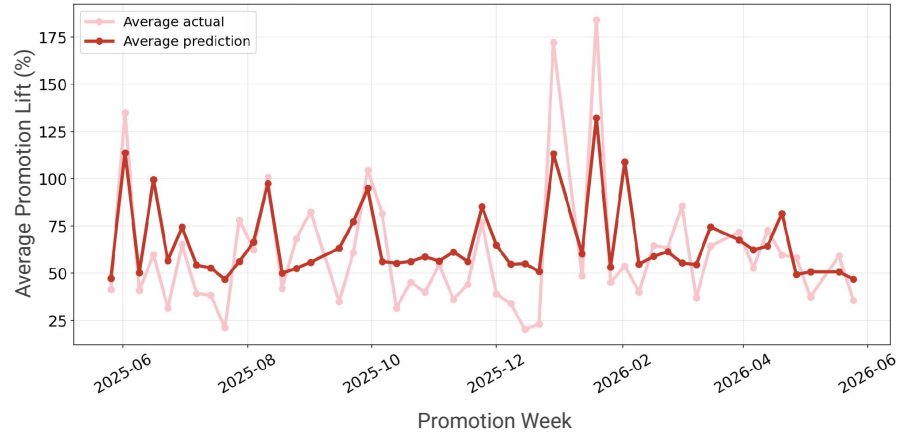
Brand Example: Coca-Cola

Weekly Lift: Actual vs. Prediction



Account Example: Customer A

Weekly Lift: Actual vs. Prediction



► The Model Delivers Significant Financial Upside

Model Coverage

7k

Stores

25%

Total Orders with
Promotion

Revenue Captured

+400k

USD per week

Compared with current
ordering behavior

+677k

USD per week

Compared with output
of Suggested Order

► Implementation Plan

Pilot Testing

Compare three ordering approaches in a controlled pilot



Current Replenisher Practice



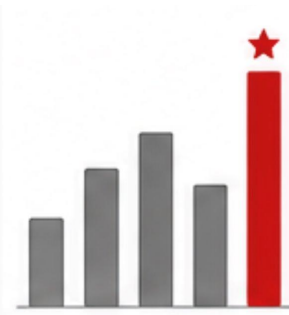
Current Suggested Order



Suggested Order + Lift

Promotion Planning

Use predicted lift to optimize promotion planning



Promotion	Predicted Lift
Promo A	45% ★★★★★
Promo B	32% ★★★★★
Promo C	18% ★★★★★
Promo D	8% ★★★★★



**Huge thanks to
MBAn and CCSWB!**

